

Forward and backward span for verbal and visuo-spatial data: standardization and normative data from an Italian adult population

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Abstract The Digit span and Corsi span tasks are frequently used to assess verbal and visuo-spatial short-term memory. Forward versions of these tasks, in which sequences of items of increasing length have to be reproduced in the order they were presented, are believed to primarily evaluate the functioning of the working memory material-specific slave systems (i.e. the phonological loop and the visuo-spatial sketchpad for verbal and visuo-spatial data, respectively). The backward versions of both tasks, in which sequences of items have to be reproduced in the reverse order, are believed to primarily tax Central Executive resources. Here, we report normative data on the forward and backward versions of the Digit span and Corsi span tasks that was collected from 362 healthy Italians ranging in age from 20 to 90 years. The results show a decremental effect of age on performance in all tasks and an ameliorative effect of education in all tasks except the Corsi span backwards. We provide correction grids for age and literacy that derive from results of the regression analyses.

Keywords Span · Verbal · Visuo-spatial · Working memory

Introduction

Complex cognitive tasks, such as language comprehension, mental arithmetic and reasoning, require that relevant information (either externally derived or internally generated) be transitorily kept in mind and opportunely manipulated for the specific goals of the task. The cognitive function underlying transient maintenance and online processing of information is known as working memory.

According to Baddeley's [1, 2] influential model, working memory consists of two main subcomponents: the first, namely, the Central Executive, is involved in focusing attention on relevant information, suppressing irrelevant stimuli and inappropriate actions, and coordinating multiple cognitive processes when more than one task has to be performed simultaneously; the second main subcomponent consists of a number of Slave Systems, each responsible for the short-term maintenance of a specific kind of information. The most thoroughly investigated Slave Systems are the phonological loop, which stores verbal information and prevents its decay by articulating its contents in a rehearsal loop [3], and the visuo-spatial sketchpad, which is involved in the storage of both the shapes, visual and spatial location of visual information by means of partially distinct cognitive and neural mechanisms [4]. Different neural substrates underlie the Central Executive and the two Slave Systems. Indeed, the prefrontal areas of both hemispheres are believed to be involved in the functioning of the Central Executive [5], and distinct circuitries, including the temporal, parietal and frontal lobes of the right and left hemisphere, are believed to underlie the functioning of the phonological loop and the visuo-spatial sketchpad, respectively [6, 7].

In clinical neuropsychological practice, the most widely used tests of verbal and visuo-spatial working memory are the Digit span [8–10] and Corsi block tasks [11, 12]. In the

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original versions (Digit span and Corsi block forward), patients are presented with sequences of verbal (digits) or spatial (block positions) items they have to reproduce immediately after presentation in the same temporal order. The length of the sequence is progressively increased and the span (considered a measure of short-term memory store capacity) is the longest sequence correctly reproduced. In the backward versions of both the Digit span and Corsi block tasks, subjects are required to reproduce the presented sequences in reverse order. According to Baddeley's working memory model, the forward versions of the two tasks investigate short-term maintenance of verbal and visuo-spatial information without requiring further mental manipulation of the stored data. Therefore, these tasks might be adequately managed by the Slave Systems dedicated to the retention of verbal and visual data (phonological loop and visuo-spatial sketchpad, respectively) and might not involve, or only minimally involve, Central Executive resources. Conversely, the backward tasks not only require maintaining the items, but also mentally manipulating them to achieve the correct reversed reproduction. Therefore, to a larger extent these tasks engage resources of the Central Executive [5].

Previous authors collected data from large samples of Italian population to provide norms for the forward version of the Digit span [10, 12, 13] and of the Corsi block tests [11, 12]. However, normative data on the backward version of the Digit span and Corsi block test are currently unavailable. Here, we report the results of a study, which included 362 healthy individuals ranging in age from 20 to 90 years that was aimed at providing Italian norms for both the forward and backward versions of these two tests in the same sample. Our purpose was to provide a practical tool that requires little time for analyzing the functioning of the phonological loop (Digit span forward), visuo-spatial sketchpad (Corsi block forward) and Central Executive (backward versions of both tests) components of the working memory system.

Materials and methods

Subjects

We enrolled 362 healthy Italians (187 males: mean age = 53.56 ± 27.27 years, range 20–90; mean education

= 11.63 ± 4.61 years, range 2–19; 175 females: mean age = 54.82 ± 20.62 years, range 20–90; mean education = 10.93 ± 4.64 , range 2–19), coming mainly from centre and south areas of Italy. For each of seven decades (31–40, 41–50, 51–60, 71–80, and 81–90), we enrolled 25 males and 25 females; the 20–30 decade included 33 males and 25 females and the 61–70 decade included 29 males and 25 females.

The only inclusion criterion for participants in the normative group was age between 20 and 90 years. Exclusion criteria included: (1) suspected cognitive impairment or dementia based on a Mini Mental State Examination (MMSE) [14] score ≤ 26 , (2) subjective complaint of memory difficulties or any other cognitive deficits interfering or not with daily living activities, (3) reported psychiatric or neurological disorders, (4) known or suspected history of alcoholism or drug dependence or abuse during the lifetime. Table 1 reports the number of patients, distinguished by gender, for each of the seven decades considered.

Local ethics committee approval was received, and written informed consent was obtained from each participant before the study began.

Materials and procedure

The procedure for the *Digit span forward task* was that originally proposed by Hebb [15]. The examiner pronounces a list of digits at a rate of approximately one digit per second and subjects are required to immediately repeat the list in the same order. If they succeed, a list one digit longer is presented. If they fail, a second list of the same length is presented. If subjects are successful on the second list, a list one digit longer is given, as before. However, if subjects also fail on the second list, the test is ended. The length of the digit sequences gradually increase, starting with a sequence of three numbers (e.g., 5, 8, 2) to a sequence of maximum nine items (e.g., 7, 1, 3, 9, 4, 2, 5, 6, 8). The span is established as the length of the longest list recalled correctly. The same procedure is used for the *Digit span backward task*, except that in this case subjects have to reproduce the sequence of digits in the reverse order and the longest list consists of eight items.

The *Corsi block forward task* was administered according to the procedure described in Milner's [9] study.

Table 1 Experimental sample used for calculating normative data on the forward and backward versions of the Digit and Corsi span tasks

Decade	20–30	31–40	41–50	51–60	61–70	71–80	81–90
Number	50	58	50	50	54	50	50
Mean (SD) years of education (range 3–17)	12.9 (3.3)	13.8 (3.7)	12.5 (4.0)	11.8 (4.4)	10.7 (4.7)	10.2 (4.7)	6.8 (4.1)
M/F	25/25	33/25	25/25	25/25	29/25	25/25	25/25

The experimental apparatus consists of a board (25 × 35 cm) on which nine cubic blocks (side cm 4) are fixed. A serial number (from 1 to 9) is located on one side of each block and visible only to the examiner. The test starts with the examiner touching a sequence of three blocks at a rate of approximately one block per second, according to the numerical sequence that corresponds with the numbers on each block. Immediately following the presentation of the sequence, the patient has to touch the blocks in the same order. If he succeeds, the examiner touches another sequence of blocks containing an additional item. If the subject fails, a second sequence of the same length is given and, if the subject succeeds, a sequence one block longer is given. When the subject fails on two consecutive sequences of the same length the test is ended. The span is established as the length of the longest sequence correctly reproduced. The same procedure was followed for the *Corsi block backward task*. The only difference was that the subjects had to reproduce the sequence of blocks in the reverse order and the longest list included eight items.

Table 2 reports the sequences used for the forward and backward versions of both the Digit span and the Corsi block tasks.

Statistical analyses

The statistical analysis was aimed at calculating the effect of demographic variables (age, education and gender) on healthy individuals' performance of the cognitive tasks and obtaining adjusted performance

scores, that is, scores corrected for the influence of individual characteristics on demographic variables. We considered the length of the longest sequence correctly reproduced in each of the four tasks (Digit span and Corsi block, forward and backward) as the main performance index. Moreover, to obtain an estimate of the performance decrease passing from reproduction of the forward to the backward sequence, we calculated the ratio between the adjusted performance scores on the backward over the forward task for both the Digit span and the Corsi block tasks.

The effect of each demographic variable on the performance indexes was first assessed singly. As for continuous variables (age and education), correlation coefficients were calculated between the test scores and the raw and transformed values of the variable of interest. Consistent with the previous work [11], age was converted into a logarithm of 100-age and education into a square root. The effect of gender on test scores was determined using Student's *t* test and contrasting the performance of males and females. Then, for each performance score, a multiple linear regression analysis was performed with test score as the dependent variable; in these analyses, age and education (or their transformations) and gender (dummy variable) were inserted as independent variables only if they had been significant ($p < 0.05$) in the previous (single) analysis. The independent variables were entered in the definitive regression model only if their significance level was at least $p = 0.025$ ($p = 0.05/2$). Based on the results of multiple regression analyses, regression equations were built up that allowed calculating the adjusted performance scores for each combination of significant demographic variables. Once adjusted scores for each of healthy participants had been calculated, the lower limit of the tolerance interval for each of the performance indexes and the backward/forward ratios was determined according to Capitani [12]. For this purpose, adjusted scores and ratios were ranked. Then, using a nonparametric procedure [16] we set the tenth scalar observation as the tolerance limit, leaving above at least 95 % of the normal population (with 95 % confidence). A score below this tolerance limit was declared "not normal" with 95 % confidence. Conversely, a score above the tolerance limit was declared "normal" with 95 % confidence. To make scores on the span tasks comparable with those obtained on other neuropsychological tests, we classified adjusted scores into five categories (Equivalent Scores) endowed with an ordinal relationship: 0 = scores lower than the 95 % tolerance limit; 4 = scores higher than the median value of the sample; 1, 2 and 3 = scores lower than 10.4, 26.4 and 50 % of the normative sample distribution, respectively. A more detailed explanation of the equivalent scoring system is provided by Capitani [12].

Table 2 Sequences used in the forward and backward versions of the Digit and Corsi span tasks

Forward	Backward
Sequences	
5, 8, 2	6, 2, 9
6, 9, 4	4, 1, 5
6, 4, 3, 9	3, 2, 7, 9
7, 2, 8, 6	1, 9, 6, 8
4, 2, 7, 3, 1	1, 5, 2, 8, 6
7, 5, 8, 3, 6	6, 1, 8, 4, 3
6, 1, 9, 4, 7, 2	5, 3, 9, 4, 1, 8
3, 9, 2, 4, 8, 7	7, 2, 4, 8, 5, 6
5, 9, 1, 7, 4, 2, 8	8, 1, 2, 9, 3, 6, 5
4, 1, 7, 9, 3, 8, 6	4, 7, 3, 9, 1, 2, 8
5, 8, 1, 9, 2, 6, 4, 7	9, 4, 3, 7, 6, 2, 5, 6
3, 8, 2, 9, 5, 1, 7, 4	7, 2, 8, 1, 9, 6, 5, 2
2, 7, 5, 8, 6, 2, 5, 8, 4	
7, 1, 3, 9, 4, 2, 5, 6, 8	

Results

On an average, the forward span was larger in the Digit than the Corsi task (5.84 ± 1.09 vs. 5.39 ± 1.09 ; $F = 52.4$; $p < 0.0001$ with 1,361 *df*). Conversely, the backward span was larger on the Corsi than the Digit task (4.47 ± 1.19 vs. 4.72 ± 1.12 ; $F = 13.5$; $p = 0.0003$ with 1,361 *df*).

As shown in Fig. 1, which reports the means of the healthy participants according to decade, in all tasks performance was negatively associated with age. In fact, a significant correlation emerged between age and performance scores, with the correlation coefficient slightly higher for the logarithmic transformation (100 age) in all cases: Digit span forward ($r = 0.47$; $p < 0.001$) and backward ($r = 0.45$; $p < 0.001$), Corsi block forward ($r = 0.47$; $p < 0.001$) and backward ($r = 0.50$; $p < 0.001$). The correlation between performance scores and education was consistently higher following the square root transformation of years of schooling: Digit span forward ($r = 0.37$; $p < 0.001$) and backward ($r = 0.41$; $p < 0.001$), Corsi block forward ($r = 0.40$; $p < 0.001$) and backward ($r = 0.27$; $p < 0.001$). Finally, a gender effect emerged only on the Corsi block backward, with males (4.83 ± 1.07) performing significantly better than females (4.19 ± 1.17) ($F = 14.5$; $p < 0.04$ with 1,360 *df*).

Table 3 reports regression equations and correction grids by age and education for performance scores on the forward and backward versions of the Digit span and Corsi block tasks. As can be seen, transformed values of age and education consistently affected performance on the forward and backward versions of the Digit span task and the forward Corsi task. Instead, only age affected performance on the Corsi backward task.

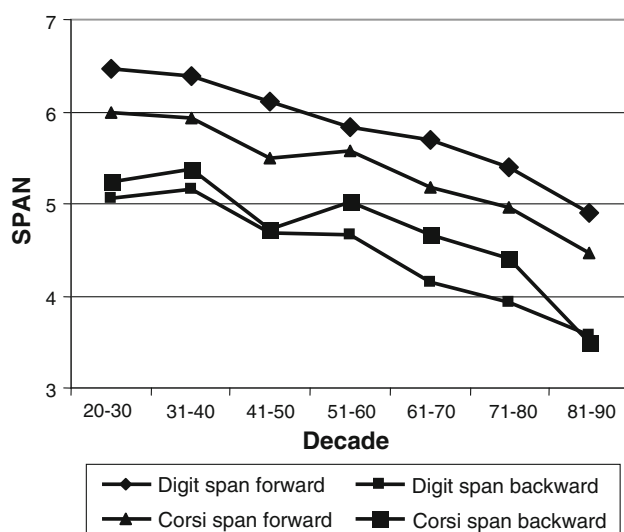


Fig. 1 Average span of healthy individuals from different decades on the forward and backward versions of the Digit and Corsi span tasks

Table 4 reports equivalent scores for performance on the forward and backward versions of both the Digit span and Corsi block tasks and for the ratio between the scores on the backward and forward versions of each task.

Discussion

The aim of this study was to provide standardization and normative data for the forward and backward versions of the Digit and Corsi span tasks. For this purpose, the performance data of a relatively large group of healthy individuals were analyzed with respect to socio-demographic variables, such as age, sex and education. As expected, increasing age was associated with progressively lower span scores on all span tasks. Instead, high education predicted high span scores on all, but the backward Corsi task. Finally, gender did not significantly affect performance on any span task. Indeed, the significantly better performance of male than female participants on the Corsi backward task in the univariate analysis was not confirmed in the multiple regression model.

With respect to the normative data reported by Orsini et al. [12] on forward tasks, our sample of healthy participants on average performed better on both Digit span (5.84 ± 1.08 vs. 5.3 ± 1.1) and Corsi span (5.38 ± 1.09 vs. 4.55 ± 0.8) tasks. This was probably due to educational differences; in fact, our sample of healthy participants was, on an average, significantly more educated than Orsini et al.'s [12] normative sample (11.30 ± 4.63 vs. 9.44 ± 5.52 years of schooling). Moreover, Orsini et al. [12] reported an effect of age and education on both the Digit and Corsi forward span tasks and an effect of gender on the Corsi task. However, as noted above, we did not observe a significant effect of gender in any of our span tasks. A direct comparison between our data and findings from WAIS-R [13] sample is substantially unwarranted because they reported a unique score expression of the cumulative forward and backward span.

The decrement in span size passing from the forward to the backward version of the task was, on an average, significantly larger for the Digit than the Corsi task (-1.36 ± 1.04 vs. -0.76 ± 1.17). This is likely due to differences in the way the two slave systems operate to transitorily maintain verbal and visuo-spatial data. Indeed, according to Baddeley [1, 17] the phonological loop maintains the speech-based material by means of articulatory rehearsal of the contents of the phonological buffer. This mechanism is well suited for preserving incoming material in the same temporal order it was presented. Therefore, when the verbal material has to be recalled backwards, performance deteriorates considerably. The mechanism by which the visuo-spatial sketchpad keeps

Table 3 Regression equations and correction grids according to age and education for performance scores on the forward and backward versions of the Digit span and Corsi block tasks

Age		20–30			31–40		41–50		51–60		61–70		71–80		81–90		Total
Digit span forward																	
Mean		6.47			6.38		6.12		5.80		5.70		5.39		4.92		5.84
SD		0.94			1.09		1.15		0.95		0.92		0.86		0.81		1.09
EVA	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90		
3	−0.09	−0.04	0.01	0.07	0.13	0.19	0.26	0.34	0.43	0.53	0.65	0.79	0.96	1.17	1.48		
5	−0.23	−0.18	−0.13	−0.07	−0.01	0.05	0.12	0.20	0.29	0.39	0.51	0.65	0.82	1.03	1.34		
8	−0.39	−0.35	−0.29	−0.24	−0.18	−0.11	−0.04	0.04	0.13	0.23	0.35	0.49	0.65	0.87	1.18		
13	−0.61	−0.56	−0.51	−0.45	−0.39	−0.32	−0.25	−0.17	−0.08	0.02	0.13	0.27	0.44	0.66	0.96		
17	*	−0.70	−0.65	−0.59	−0.53	−0.47	−0.40	−0.32	−0.23	−0.13	−0.01	0.13	0.30	0.51	0.82		
Age		20–30			31–40		41–50		51–60		61–70		71–80		81–90		Total
Digit span backward																	
Mean		5.07			5.16		4.68		4.66		4.15		3.92		3.60		4.47
SD		1.25			1.04		1.04		1.24		0.91		0.98		0.67		1.19
EVA	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90		
3	0.14	0.19	0.24	0.29	0.35	0.41	0.48	0.55	0.64	0.73	0.84	0.97	1.13	1.34	1.63		
5	−0.06	−0.02	0.03	0.08	0.14	0.20	0.27	0.35	0.53	0.53	0.64	0.77	0.93	1.13	1.42		
8	−0.31	−0.26	−0.21	−0.16	−0.10	−0.04	0.02	0.10	0.19	0.28	0.39	0.52	0.68	0.89	1.18		
13	−0.62	−0.58	−0.53	−0.48	−0.42	−0.36	−0.29	−0.21	−0.13	−0.03	0.08	0.21	0.37	0.57	0.86		
17	*	−0.79	−0.74	−0.69	−0.63	−0.57	−0.50	−0.42	−0.34	−0.24	−0.13	0.00	0.16	0.36	0.65		
Age		20–30			31–40		41–50		51–60		61–70		71–80		81–90		Total
Corsi span forward																	
Mean		6.00			5.94		5.50		5.56		5.17		5.02		4.42		5.38
SD		1.09			1.06		1.02		1.02		0.98		0.75		0.89		1.09
EVA	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90		
3	0.00	0.05	0.10	0.16	0.22	0.28	0.35	0.43	0.51	0.61	0.72	0.86	1.02	1.23	1.53		
5	−0.16	−0.11	−0.06	−0.01	0.05	0.11	0.18	0.26	0.35	0.44	0.56	0.69	0.85	1.07	1.36		
8	−0.35	−0.31	−0.26	−0.20	−0.14	−0.08	−0.01	0.07	0.15	0.25	0.36	0.50	0.66	0.87	1.17		
13	−0.61	−0.56	−0.51	−0.46	−0.40	−0.33	−0.26	−0.19	−0.10	0.00	0.11	0.24	0.41	0.62	0.91		
17	*	−0.73	−0.68	−0.63	−0.57	−0.50	−0.43	−0.36	−0.27	−0.17	−0.06	0.07	0.24	0.45	0.74		
Age		20–30			31–40		41–50		51–60		61–70		71–80		81–90		Total
Corsi span backward																	
Mean		5.24			5.38		4.70		5.04		4.66		4.43		3.50		4.72
SD		0.90			1.14		0.91		1.05		0.95		0.84		1.00		1.13
Age	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90		
	−0.83	−0.76	−0.69	−0.61	−0.52	−0.43	−0.33	−0.21	−0.09	0.06	0.22	0.42	0.66	0.96	1.40		

Best linear model: $5.84 + 1.74 * (\log(100 - \text{age}) - 1.61) + 0.28 * (\sqrt{\text{school}} - 3.28)$

Significance: age $t = 7.19$, $p < 0.025$; school $t = 3.60$, $p < 0.025$

Best linear model: $4.47 + 1.65 * (\log(100 - \text{age}) - 1.61) + 0.41 * (\sqrt{\text{school}} - 3.28)$

Significance: age $t = 6.23$, $p < 0.025$; school $t = 4.88$, $p < 0.025$

Best linear model: $5.38 + 1.69 * (\log(100 - \text{age}) - 1.61) + 0.33 * (\sqrt{\text{school}} - 3.28)$

Significance: age $t = 6.98$, $p < 0.025$; school $t = 4.28$, $p < 0.025$

Best linear model: $4.72 + 2.41 * (\log(100 - \text{age}) - 1.61)$

Significance: age $t = 11.04$, $p < 0.025$

Table 4 Equivalent scores for performance on the forward and backward versions of both the Digit span and Corsi block tasks and for the ratio between the scores on the backward and forward versions of each task

Equivalent scores	0 2.90 %	1 10.40 %	2 26.40 %	3 50 %	4 100 %
Percentage of sample distribution below the reported value					
Digit span forward	<4.26	<4.60	<5.29	<5.75	>5.75
Digit span backward	<2.65	<3.29	<3.79	<4.33	>4.33
Corsi span forward	<3.46	<4.29	<4.80	<5.37	>5.37
Corsi span backward	<3.08	<3.37	<3.99	<4.55	>4.55
Ratio digit span	<0.48	<0.58	<0.66	<0.79	>0.79
Ratio Corsi span	<0.51	<0.63	<0.74	<0.90	>0.90

spatial data active is less known, but may be related to attentional processes operating over the contents of a representational buffer [18]. Conceivably, it is less difficult for this system to report the spatial sequence in a different temporal order than it was initially presented in, so that the decrement in accuracy is less severe passing from the forward to the backward span task.

Our normative data can be used in the clinical evaluation of the functioning of the phonological loop and visuo-spatial sketchpad. Damage to these components of the working memory system can be observed in focal brain pathologies [19] and degenerative brain diseases [20], which affect the dedicated neural circuitries. Moreover, an indirect estimate of the contribution of the Central Executive in the short-term maintenance of verbal and visuo-spatial data can be derived from the ratio between performance on the backward over the forward task on both the Digit span and the Corsi block tasks. Indeed, as noted above, the processing resources of the Central Executive are required to actively maintain and manipulate the verbal and spatial sequences to be reported in reverse order. Very low ratio scores are indicative of a particularly large performance decrement passing from the forward to the backward span tasks; in turn, this could be related to the impoverishment of Central Executive resources, as observed in frontal lobe focal and degenerative pathologies [21, 22].

Application of our normative data to the performance of groups of brain-damaged patients on verbal and span tasks should help clarify the clinical usefulness of this very popular and easy to administer instrument for assessing short-term memory disorders and their impact on the functioning of other, higher order, cognitive abilities (e.g. long-term declarative memory, reading comprehension and visual imagery).

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